EgoReasoner: Egocentric Embodied Reasoning with MLLMs via Reinforcement Learning

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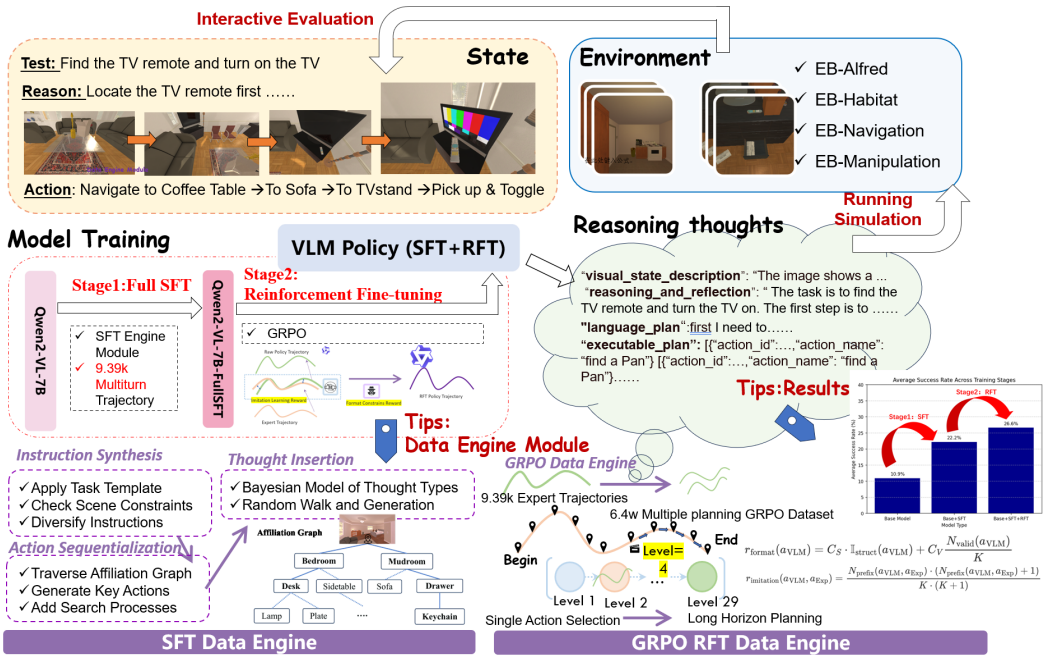


Figure 1: Overview of the EgoReasoner framework.

Abstract

Multimodal Large Language Models (MLLMs) struggle with embodied reasoning tasks requiring long-horizon planning, temporal logic, and spatial understanding. We introduce EgoReasoner, a framework that enhances MLLM planning capabilities through multi-stage training. We first synthesize expert trajectories using hybrid rule-based logic and LLM calls, then apply Supervised Fine-Tuning (SFT) to instill foundational planning abilities. We further refine decision-making via Group Relative Policy Optimization (GRPO), which leverages relative preferences between trajectory groups. On EmbodiedBench tasks, our approach improves success rates from 10.9% (zero-shot) to 22.2% (SFT) and 26.6% (GRPO), demon-

strating that compact MLLMs can achieve strong embodied reasoning through structured supervision and reinforcement learning.

1 Introduction

While Multimodal Large Language Models (MLLMs) excel at language grounding and visual understanding, they struggle with long-horizon embodied tasks [1, 2]. Temporal reasoning, spatial understanding, and multi-step planning remain challenging [3, 4], motivating our exploration of efficient enhancement frameworks.

We propose a lightweight training pipeline for embodied MLLM agents that combines structured supervision with reinforcement learning. Our approach synthesizes high-quality expert trajectories using rule-based logic and LLM-generated actions [5, 6], which seed Supervised Fine-Tuning (SFT) to establish behavioral priors.

For long-horizon reasoning refinement [7, 8], we employ Group Relative Policy Optimization (GRPO) [9]. GRPO enhances policy learning through relative trajectory comparisons, providing more stable optimization than absolute rewards. Validated on interactive embodied search tasks, our framework demonstrates that even compact MLLMs can become competent embodied agents through high-quality synthetic data, supervised pre-training, and reinforcement-based fine-tuning.

2 Related Work

2.1 Deep Thinking and Reasoning Models

The emergence of deep-thinking models such as OpenAI o1, DeepSeek-R1, and related systems has revolutionized complex reasoning tasks, particularly in mathematical and coding domains. These models employ a "slow-thinking" paradigm where they generate extensive reasoning chains before producing final answers, achieved through sophisticated reinforcement learning training on reasoning trajectories. However, extending this paradigm to embodied scenarios presents unique challenges. Unlike mathematical reasoning that relies primarily on logical deduction, embodied tasks require spatial understanding, temporal reasoning across interaction histories, and continuous adaptation to environmental feedback. Recent work by Zhao et al. [2] introduces Embodied-R, which addresses computational constraints by decoupling perception and reasoning through a collaborative framework combining large-scale vision-language models for perception with smaller language models for reasoning, trained via reinforcement learning with novel logical consistency rewards.

2.2 Tool Use and Interactive Clarification

A critical aspect of embodied reasoning involves effective tool use and handling of ambiguous or incomplete user intents. Traditional tool learning frameworks assume explicit and unambiguous queries, which diverges from real-world scenarios where users often provide incomplete or imprecise instructions. Zhang et al. [10] address this challenge through ASKTOACT, a self-correcting clarification framework that leverages the structural mapping between queries and tool invocation solutions. Their approach generates high-quality training data by systematically removing key parameters from complete queries while retaining them as ground truth, enabling automated construction of clarification dialogues. The framework incorporates error-correction mechanisms that allow models to detect and recover from mistakes during multi-turn interactions, achieving significant improvements in intent recovery and clarification efficiency.

2.3 Embodied Spatial Reasoning

Spatial reasoning represents a fundamental challenge in embodied AI, requiring models to perceive and reason about spatial relationships from sequential visual observations. Current approaches often struggle with the complexity of spatio-temporal relationships in video data and the distinct characteristics of egocentric visual observations. The Embodied-R framework [2] tackles these challenges through a collaborative architecture that separates perception and reasoning components, enabling efficient processing of continuous visual streams while maintaining computational tractability. Their

approach demonstrates that small-scale language models can achieve comparable performance to much larger multimodal reasoning models when properly trained with reinforcement learning techniques that emphasize logical consistency.

2.4 Training Paradigms for Embodied Agents

The development of effective embodied agents requires sophisticated training paradigms that can handle the complexity of interactive environments. Recent approaches have explored various combinations of supervised fine-tuning (SFT) and reinforcement learning (RL) to enhance agent capabilities. Zhang et al. [1] present a comprehensive three-stage training pipeline that progressively builds agent capabilities through imitation learning, self-exploration via rejection sampling, and self-correction through reflection tuning. This iterative approach addresses the challenge of learning from both expert demonstrations and self-generated experience, enabling models to develop robust reasoning patterns for long-horizon tasks.

Our work builds upon these foundations by proposing a lightweight yet effective pipeline that combines high-quality synthetic data generation with a hybrid training framework incorporating both SFT and GRPO. This approach addresses the computational efficiency concerns raised by previous work while maintaining the sophisticated reasoning capabilities demonstrated by deep-thinking models, specifically tailored for long-horizon embodied interactive tasks.

3 Methods

We formulate embodied reasoning as sequential decision-making. At timestep t, the agent observes $o_t \in \mathcal{O}$ and executes action $a_t \in \mathcal{A}$ via policy π_θ (an MLLM). The interaction history is $h_t = \{o_0, a_0, \ldots, o_t\}$. Given goal g and task prompt L, trajectory $\tau = \{g, o_0, a_0, \ldots, o_n, a_n\}$ is generated. Our objective: train π_θ to complete tasks successfully (Figure 1).

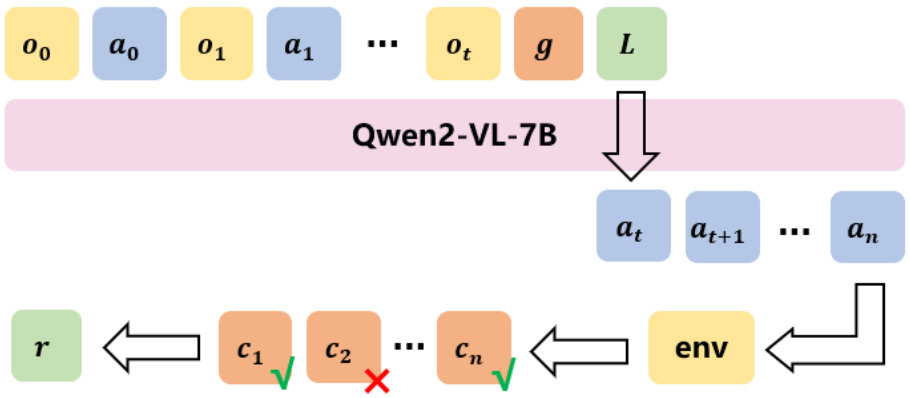


Figure 2: Model architecture overview.

EgoReasoner fine-tunes an MLLM (Qwen-VL) via imitation learning (SFT) and reinforcement learning (GRPO).

3.1 Trajectory Synthesis and Supervised Fine-Tuning

We bootstrap training with expert-like demonstrations from a hybrid system combining rule-based logic and LLM calls, avoiding expensive human annotation while ensuring quality and diversity.

The synthesis process comprises three steps:

- **Instruction Synthesis:** We define four task templates (search, manipulation, transportation, composition), sample template-object pairs, verify scene consistency and tractability, then use LLMs to diversify instruction styles and difficulties.

- **Action Sequentialization:** We construct an affiliation graph from scene object co-relations, then traverse it instruction-guided to generate key action sequences [1, 2].
- **Thought Insertion:** Following Bayesian models of human reasoning [11], we use Monte Carlo sampling on a probabilistic graph to generate five thought types (analysis, planning, reflection, spatial reasoning, verification) via LLMs.

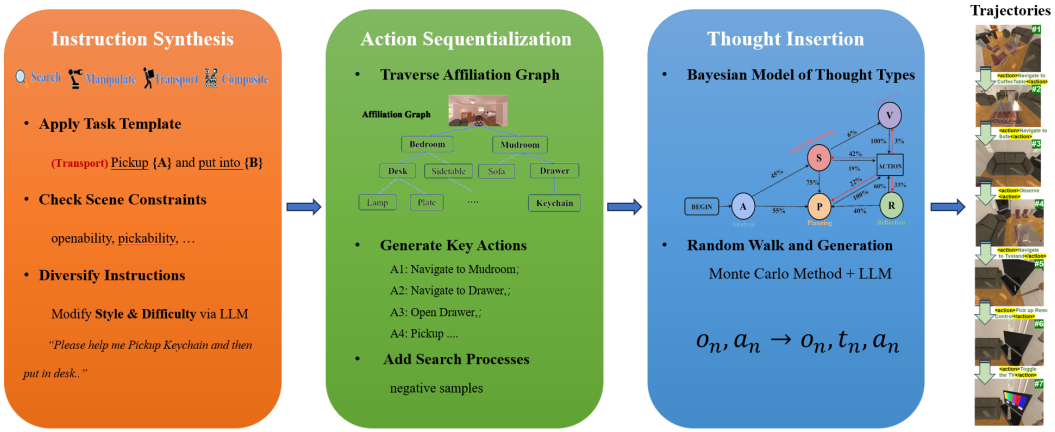


Figure 3: SFT data generation engine.

We collect ∼10K synthesized trajectories for SFT on Qwen2-VL-Instruct-7B, teaching structured reasoning patterns and proper action formatting (`<Think>...<\Think><Act>...<\Act>`). This provides crucial initialization for reinforcement learning. We use the Swift framework for SFT.

3.2 Group Relative Policy Optimization (GRPO)

Post-SFT, we enhance reasoning via GRPO, a preference-based RL algorithm that compares trajectory groups. For each task, we generate multiple candidate trajectories, evaluate and rank them, then update policy using relative preferences. This group-wise comparison provides more robust signals than pairwise comparisons or absolute rewards [9].

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(O|q)]$$
$$\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\left\{\min\left[\frac{\pi_\theta(o_{i,t}|q,o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q,o_{i,<t})}\hat{A}_{i,t}, \text{clip}\left(\frac{\pi_\theta(o_{i,t}|q,o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q,o_{i,<t})}, 1-\varepsilon, 1+\varepsilon\right)\hat{A}_{i,t}\right] - \beta\mathbb{D}_{KL}\left[\pi_\theta||\pi_{ref}\right]\right\},$$

Figure 4: GRPO optimization objective.

3.2.1 GRPO with Imitation Learning (Offline)

We leverage expert trajectories from synthesis for offline GRPO. For reference sequence length k with generated prefix matching n steps, multi-step accuracy reward is:

$$R_{\text{accuracy}} = \frac{n(n+1)}{k(k+1)}$$

This encourages correct action sequences beyond single-step accuracy. Format rewards penalize template deviations. Final reward combines these components, enabling fine-tuning without live simulation.

3.2.2 GRPO with Online Simulation

For error recovery and strategy exploration, we implement online GRPO. Each training step executes the agent's action sequence in a simulator instance. Total reward R_{sim} comprises:

- **Format Reward**: Penalizes incorrect output formatting.
- **Task Completion Reward**: Large positive for success, small penalty for failure.
- **Length Penalty**: Discourages overly long sequences.
- **Repetition Penalty**: Penalizes oscillatory behaviors.

4 Experiments and Results

4.1 Experimental Setup

Benchmarks We evaluate on EmbodiedBench [12] subsets:

- **EB-ALFRED**: Multi-step interactive tasks in household environments.
- **EB-Habitat**: Complex indoor navigation and interaction.

These benchmarks test reasoning, planning, and perception in visually rich interactive settings.

Baselines We compare EgoReasoner-7B against Qwen2-VL-7B-Instruct, Qwen2-VL-72B-Instruct, Qwen-VL-Max, and Embodied-Reasoner-7B [**embodied-reasoner**].

4.2 Performance Evaluation

Table 1 shows stage-wise improvements. Our 7B model achieves 10.9% zero-shot, doubles to 22.2% post-SFT, and reaches 26.6% after GRPO, validating each component.

Table 1: Stage-wise performance of EgoReasoner-7B. Success rates (%) on held-out embodied search tasks.

Model Stage	Success Rate (%)
Zero-shot	10.9
After SFT	22.2
After GRPO	26.6

Table 2 compares our model against state-of-the-art MLLMs. EgoReasoner-7B significantly outperforms similar-sized models and rivals much larger ones. On EB-ALFRED, we achieve 40.1%, exceeding Embodied-Reasoner-7B by 1.6 points, demonstrating our framework's effectiveness. Figure 5 illustrates qualitative reasoning.

Table 2: Model success rates (%) on EB-ALFRED and EB-Habitat.

Model	EB-ALFRED	EB-Habitat
Qwen2-VL-7B-Ins	1.7	18.3
Embodied-Reasoner-7B [**embodied-reasoner**]	38.5	41.1
EgoReasoner-7B (Ours)	**40.1**	**41.5**
Qwen2-VL-72B-Ins	33.7	35.7
Qwen-VL-Max (∼235B)	41.3	45.3

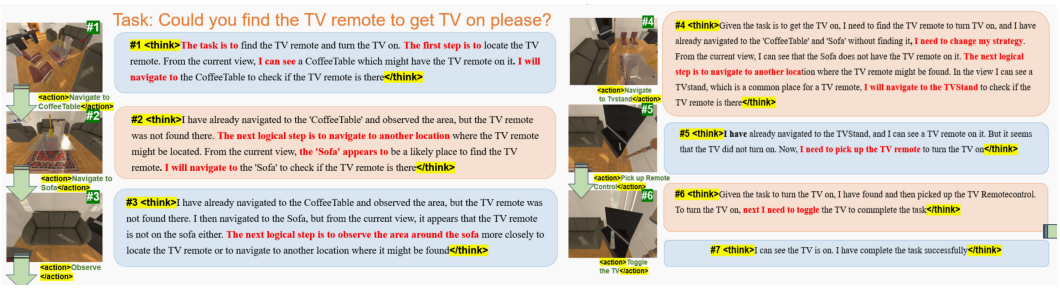


Figure 5: EgoReasoner's interactive embodied reasoning: observation, reasoning, and action execution.

5 Conclusion and Discussion

EgoReasoner demonstrates that combining Supervised Fine-Tuning on synthetic data with Group Relative Policy Optimization significantly enhances compact Vision-Language Models for long-horizon embodied tasks. Our 7B model rivals and surpasses larger baselines, validating this methodology for efficient embodied agents. However, critical examination of VLM planning capabilities and reflection on paths toward general embodied intelligence is essential.

5.1 Planning Emergence and Mechanisms

EgoReasoner's success stems from staged training. SFT establishes *behavioral priors*, constraining output space from general language generation to goal-oriented policy with task syntax (`<Think>...<Act>...`), preventing inefficient exploration during RL.

GRPO refines decision-making through relative trajectory ranking rather than noisy absolute rewards, excelling where dense reward functions are intractable. Online simulation enables learning from mistakes beyond imitation, fostering robust, generalizable strategies that explain post-GRPO performance gains.

5.2 Capability Boundaries and Limitations

Despite progress, fundamental limitations persist.

Physical Commonsense and Causal Reasoning. Current "reasoning" is statistical pattern matching, not causal physical understanding. The model learns not to traverse walls from trajectory absence, not solidity comprehension. Knowledge is correlational, not causal. Novel scenarios (e.g., permeable barriers) would likely fail, lacking first-principles grounding. It knows *what* works in familiar contexts, not *why*.

Open-World Generalization. Performance on EB-ALFRED and EB-Habitat benchmarks is strong, but these remain closed-world systems with finite objects, layouts, and affordances. Generalization to truly open environments—novel objects, unforeseen challenges, different dynamics—remains uncertain. Moving beyond fixed action sets toward *skill discovery* is crucial.

Training Efficiency and Sim-to-Real Gap. Online GRPO is computationally expensive, requiring massive parallel simulation. Research was simulation-only. The *sim-to-real gap*—discrepancy between simulation and real-world sensor noise, actuation errors, unpredictable dynamics—remains unaddressed. Habitat success doesn't guarantee physical robot performance.

5.3 Future Directions

Future research should prioritize:

Explicit World Models. Move beyond implicit models. Train VLMs to predict physical action consequences (e.g., future frames, geometric state changes). Integrate symbolic structures (knowledge graphs) for causal reasoning and planning.

Lifelong Learning. Current SFT+GRPO is offline. True autonomy requires *lifelong learning*—continuous post-deployment knowledge and policy updates from ongoing experience. Online continual learning mechanisms are essential for long-term environmental adaptation.

EgoReasoner charts an effective path for VLM-based embodied reasoning through supervised and reinforcement learning synthesis. However, achieving general-purpose embodied intelligence requires moving beyond simulated benchmarks to confront physical causality, open-world generalization, and operational safety challenges directly.

A Appendix

A.1 Predefined Action Space

The agent uses a discrete action space mapping outputs to simulator commands: navigation (`MoveAhead`, `RotateLeft`), interaction (`PickupObject`, `SliceObject`), and state changes (`OpenObject`, `CloseObject`). Figure 6 visualizes key actions.

```
{
    "navigate to <object>\": Move to the object.
    "pickup <object>\": Pick up the object.
    "put <object>\": Put the item in your hand into or on the object.
    "toggle <object>\": Switch the object on or off.
    "open <object>\": Open the object (container), and you will see inside the
object.\n\"close <object>\": Close the object.
    "observe\": You can obtain image of your directly rear, left, and right perspectives.
    "move forward\": Move forward to see more clearly.
    "end\": If you think you have completed the task, please output \"end\".
}.
```

Figure 6: Subset of predefined action space.

A.2 GRPO Training Data

Figure 7 shows GRPO training data format: prompts (visual and textual) with multiple policy-sampled responses and rewards. This structure enables group-wise comparison.

```
{
    "messages":
    [
        {
         "role": "system",
         "content": "You are a robot in given room. You need to complete the tasks
according to human instructions. We provide an Available_Actions set and the
corresponding explanations for each action. Each step, you should select one action from
Available_Actions."},
        {
         "role": "user",
         "content": "<image>This is an image from your initial frontal perspective.
Please select an action from the Available_Actions and fill in the arguments.\nTask:
\"Get a Newspaper from the room.\"\nAvailable_Actions: <Predefined Action set>\nYour
final action must strictly follow format: <DecisionMaking>Your Action</DecisionMaking>,
for example, <DecisionMaking>observe</DecisionMaking>.Before making each decision, you
can think, plan, and even reflect step by step, and then output your final
action.\n\nPrevious actions taken: navigate to CoffeeTable, navigate to CoffeeTable,
navigate to SideTable, navigate to DiningTable"},
     ],
    "action": ["pickup Newspaper", "navigate to Sofa", "pickup Newspaper"],
    "images":
["./data/images/navigate1pickup0/FloorPlan228_single_pickup_5798_fix/FloorPlan228_1_init.
png"]
}
```

Figure 7: GRPO training data instance.

A.3 Team Contributions

Zixuan Wang Project leader contributions:

- Led literature review, proposed project idea, reproduced papers, structured framework.
- Managed team roles, timelines, and experimental environment.
- Deployed baselines; developed LLM-AI2-THOR interface.
- Implemented GRPO in Swift framework, executed primary training.
- Contributed GRPO poster content.
- Wrote Experimental Results and Discussion sections.

Danqi Zhao Contributions:

- Conducted literature review, defined technical approach for spatio-temporal reasoning.
- Evaluated SFT and RL models on synthetic data and EmbodiedBench.
- Verified GRPO code, optimized training parameters, improved efficiency.
- Led poster design and formatting.

Puzhen Yuan Contributions:

- Proposed offline (imitation) and online (simulation) GRPO paradigms.
- Implemented simulation GRPO: format, length, environment rewards, training scripts.
- Debugged both GRPO variants.
- Contributed to poster and report.

Zhuo Cao Contributions:

- Conducted literature review, defined trajectory collection and SFT approach.
- Implemented data engine, collected $\sim$10K trajectories.
- Set up multi-GPU SFT framework.
- Packaged models with standard API for benchmarks.

Chenxiao Yang Contributions:

- Participated in methodology design and literature review.
- Contributed to fine-tuning: data preparation and organization.
- Collected and edited demo videos.
- Contributed to poster and report writing.

References

[1] W. Zhang et al. "Embodied-Reasoner: Synergizing Visual Search, Reasoning, and Action for Embodied Interactive Tasks". In: *arXiv preprint arXiv:2503.21696* (2025). URL: https://arxiv.org/abs/2503.21696.

[2] B. Zhao et al. "Embodied-R: Collaborative Framework for Activating Embodied Spatial Reasoning in Foundation Models via Reinforcement Learning". In: *arXiv preprint arXiv:2504.12680* (2025). URL: https://arxiv.org/abs/2504.12680.

[3] G. Wang, Y. Xie, Y. Jiang, et al. "Voyager: An Open-Ended Embodied Agent with Large Language Models". In: *arXiv preprint arXiv:2305.16291* (2023). URL: https://arxiv.org/abs/2305.16291.

[4] M. Ahn, A. Brohan, N. Brown, et al. "Do as I Can, Not as I Say: Grounding Language in Robotic Affordances". In: *arXiv preprint arXiv:2204.01691* (2022). URL: https://arxiv.org/abs/2204.01691.

[5] W. Yuan, J. Duan, V. Blukis, et al. "RoboPoint: A Vision-Language Model for Spatial Affordance Prediction for Robotics". In: *arXiv preprint arXiv:2406.10721* (2024). URL: https://arxiv.org/abs/2406.10721.

[6] R. Ramrakhya, M. Chang, X. Puig, et al. “Grounding Multimodal LLMs to Embodied Agents that Ask for Help with Reinforcement Learning”. In: *arXiv preprint arXiv:2504.00907* (2025). URL: https://arxiv.org/abs/2504.00907.

[7] J. Bai et al. “Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond”. In: *arXiv preprint arXiv:2308.12966* (2023). URL: https://arxiv.org/abs/2308.12966.

[8] J. Yang et al. “Thinking in Space: How Multimodal Large Language Models See, Remember, and Recall Spaces”. In: *arXiv preprint arXiv:2412.14171* (2024). URL: https://arxiv.org/abs/2412.14171.

[9] DeepSeek-AI et al. “DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning”. In: *arXiv preprint arXiv:2501.12948* (2025). URL: https://arxiv.org/abs/2501.12948.

[10] Xuan Zhang et al. “Asktoact: Enhancing llms tool use via self-correcting clarification”. In: *arXiv preprint arXiv:2503.01940* (2025).

[11] Yizhong Wang et al. *Self-Instruct: Aligning Language Models with Self-Generated Instructions*. 2023. arXiv: 2212.10560 [cs.CL]. URL: https://arxiv.org/abs/2212.10560.

[12] R. Yang, H. Chen, J. Zhang, et al. “EmbodiedBench: Comprehensive Benchmarking Multi-modal Large Language Models for Vision-Driven Embodied Agents”. In: *arXiv preprint arXiv:2502.09560* (2025). URL: https://arxiv.org/abs/2502.09560.